903615

Course Name: **Machine Learning**

Course Outcome

CO1-Understand the concepts of Machine learning and intelligent agents.
CO2-Understand and learn knowledge representation and reasoning for the problem-solving.
CO3-Apply basic search techniques for problem-solving.
CO4-Understand and apply learning techniques.
CO5-Apply and utilize AI knowledge for application in the real world.

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Mid Term Examination, Odd Semester 2025-26
B.Tech. CS, 3rd Year, V Semester
BCSE 0105 Machine Learning

Time: 2 Hours **Maximum Marks: 30**

Instruction for students: This paper is divided into two sections: **A** and **B.** All the sections are compulsory.

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	You are working with a dataset from a real estate company that includes features of houses such as their size, number of bedrooms, location, and age. The goal is to predict the final selling price of a house when it goes on the market. Is this a regression or classification problem? Justify your answer and explain how the model's output would differ in each case.	3	2	A	P
2	In a healthcare dataset, you observe that many rows are incomplete because the Age and Blood Pressure attributes are missing. What preprocessing methods can be used to handle these missing values? Explain when it is more appropriate to replace values with the mean, use the median, or completely drop the records.	3	2	A	C
3	In the context of spam detection, a machine learning model achieves an accuracy of 50% on the training dataset and 52% on the testing dataset. What type of modeling error does this scenario represent? Explain the nature of this error, and discuss the strategies that could be employed to enhance the model's detection performance.	3	4	U	C

4	You are given the following dataset of students and their preferred sports:
	
	import pandas as pd
	
	data = {
	'Name': ['Amit', 'Riya', 'Sohan', 'Priya', 'Raj'],
	'Sport': ['Cricket', 'Football', 'Cricket', 'Tennis', 'Football']
	}
	df = pd.DataFrame(data)
	print(df)
	
	Apply One-Hot Encoding to the Sport column using Python and represent the result in tabular form.
		3	1	A	P

5	You are given the following dataset of students with their scores in English and Maths, along with their performance label (Pass or Fail): **Student** | **English** | **Maths** | **Class** S1 | 6 | 7 | Pass S2 | 2 | 4 | Fail S3 | 7 | 8 | Pass S4 | 3 | 3 | Fail A new student, S5, has scores of English = 4 and Maths = 5. Compute the Euclidean distance of S5 from each of the four students using the K-NN classifier with k = 3, and predict the class of S5.	3	3	E	P

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	A dataset consists of the following 2D points: **(2,0), (0,2), (1,1), (3,3)** Compute the eigenvalues of the covariance matrix for this dataset using the Principal Component Analysis (PCA) algorithm and Explain the purpose of PCA in reducing high-dimensional data into lower dimensions.	5	5	A	P

7	A bank uses a Logistic Regression model to predict whether a customer is likely to default on a loan based on features like income, credit score, and employment history. The model outputs probabilities for default (1) or no default (0). Explain how Logistic Regression calculates these probabilities and what the coefficients indicate about the effect of each feature.	5	3	A	P
8	A machine learning model is developed to predict whether patients have a certain disease. The confusion matrix obtained from the test dataset is given below: **Predicted: Positive** / **Predicted: Negative** **Actual: Positive**: 70 / 30 **Actual: Negative**: 10 / 90 Based on this confusion matrix: Calculate the following evaluation metrics: (1) Accuracy, Precision, Recall (Sensitivity), TNR (Specificity), and F1-score. (2) If the model is to be used in a critical healthcare setting, which metric would be more important Precision or Recall? Justify your answer.	5	1	E	C